

# Edelstein Effect for a Rashba Fermion: Mathematical Model and Analytical Framework

## 1. Introduction and Physical Context

The **Edelstein effect (EE)**, also known as the *inverse galvanic effect*, describes the generation of a non-equilibrium spin density (magnetization) when an electric field is applied to a system with spin-orbit coupling. In a **Rashba two-dimensional electron gas (2DEG)**, the spin degeneracy of electronic bands is lifted, and the electron momentum direction is locked to its spin direction (*spin-momentum locking*). When an external electric field is applied, the Fermi surfaces are shifted, producing a non-equilibrium spin imbalance and thus a net magnetization perpendicular to both the applied field and the spin-orbit field [Edelstein, *Solid State Commun.* 73, 233 (1990); Gambardella & Miron, *Philos. Trans. R. Soc. A* 369, 3175 (2011)].

The **Rashba Hamiltonian** for a 2DEG at the  $\Gamma$  point is [Bychkov & Rashba, *JETP Lett.* 39, 78 (1984)]:

$$\hat{H} = \frac{p^2}{2m} + \alpha \hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma})$$

where:

- $m$  is the effective carrier mass,
- $\alpha$  is the Rashba spin-orbit coupling (SOC) strength,
- $\mathbf{p} = \hbar \mathbf{k}$  is the momentum,
- $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$  are the Pauli matrices,
- $\hat{z}$  is the direction perpendicular to the 2D plane.

## 2. Model Hamiltonian and Band Structure

### 2.1 Energy Dispersion

The eigenenergies of the Rashba Hamiltonian are [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\varepsilon_{\pm}(\mathbf{k}) = \frac{\hbar^2 k^2}{2m} \pm \alpha \hbar k$$

with  $k = |\mathbf{k}| = \sqrt{k_x^2 + k_y^2}$ . The two bands are labeled by the *helicity*  $\nu = \pm 1$ , defined through the helicity operator [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\hat{S} = \frac{\hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma})}{p}, \quad \text{eigenvalues } s = \pm 1$$

## 2.2 Spin Eigenstates

The spin expectation values on the eigenstates of the two bands are [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^{\pm} = \frac{1}{k} \begin{pmatrix} \pm k_y \\ \mp k_x \\ 0 \end{pmatrix} = \begin{pmatrix} \pm \sin \theta \\ \mp \cos \theta \\ 0 \end{pmatrix}$$

where  $\theta$  is the angle between  $\mathbf{k}$  and the  $x$ -axis. This shows the characteristic **tangential (in-plane) spin texture** of the Rashba system—spin is always perpendicular to the momentum.

## 2.3 Fermi Wavenumbers

The Fermi wavenumbers for each band depend on the electron density regime:

**High-Density Regime (HDR)** — both bands occupied ( $E_F > 0$  above the band crossing) [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$k_F^+ = -k_0 + \sqrt{k_0^2 + \frac{2mE_F}{\hbar^2}}, \quad k_F^- = +k_0 + \sqrt{k_0^2 + \frac{2mE_F}{\hbar^2}}$$

**Low-Density Regime (LDR)** — only the lowest band occupied ( $E_F < 0$  below the band crossing) [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$k_F^+ = +k_0 - \sqrt{k_0^2 + \frac{2mE_F}{\hbar^2}}, \quad k_F^- = +k_0 + \sqrt{k_0^2 + \frac{2mE_F}{\hbar^2}}$$

where  $k_0 = m\alpha/\hbar$ .

# 3. Boltzmann Transport Framework for the Edelstein

# Effect

## 3.1 Magnetization Formula

Within the **semiclassical Boltzmann approach**, the magnetization (total spin density) at first order in the electric field  $\mathbf{E}$  is [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\mathbf{M} = -\mu_B \sum_{\mathbf{k}, \nu} |e| \bar{\tau}_{\mathbf{k}}^{\nu} (\mathbf{v}^{\nu}(\mathbf{k}) \cdot \mathbf{E}) \delta[E^{\nu}(\mathbf{k}) - E_F] \langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^{\nu}$$

where:

- $\mu_B$  is the Bohr magneton,
- $\bar{\tau}_{\mathbf{k}}^{\nu}$  is the transport lifetime,
- $\mathbf{v}^{\nu}(\mathbf{k}) = \nabla_{\mathbf{k}} \varepsilon_{\mathbf{k}}^{\nu}$  is the group velocity,
- $e$  is the electron charge.

## 3.2 Equivalent Formulation via Perturbed Distribution Function

Following Ezawa [arXiv:2501.01888 (2025)], the magnetization can be written using the first-order correction to the Fermi distribution function:

$$f^{(1)} = \frac{e\tau}{\hbar} \mathbf{E} \cdot \nabla_{\mathbf{k}} f^{(0)}$$

giving the magnetization:

$$\mathbf{M} = \frac{e\tau}{\hbar} g\mu_B \int \frac{d^3k}{(2\pi)^3} \mathbf{S}(\mathbf{k}) f^{(1)}$$

where  $\mathbf{S}(\mathbf{k}) \equiv \langle \psi_{\mathbf{k}} | \boldsymbol{\sigma} | \psi_{\mathbf{k}} \rangle$  and  $g$  is the  $g$ -factor. At zero temperature this reduces to:

$$\mathbf{M} = e\tau g\mu_B \int \frac{d^3k}{(2\pi)^3} \mathbf{S}(\mathbf{k}) (\mathbf{E} \cdot \mathbf{v}) \delta(\varepsilon_{\mathbf{k}} - \mu)$$

## 3.3 Linear Response: Edelstein Susceptibility

The linear Edelstein effect is defined through the susceptibility tensor  $\chi_{ij}$  [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$m_j = \chi_{ij} E_i$$

or equivalently [Ezawa, arXiv:2501.01888 (2025)]:

$$M_i = e\tau \alpha_{ij}^{ME} E_j$$

with the **magnetoelectric susceptibility**:

$$\alpha_{ix}^{ME} = \frac{g\mu_B}{(2\pi)^2 W} \sum_{\pm} \int d\phi \frac{k v_x S_i(k)}{\left| \frac{\partial \varepsilon}{\partial k} \right|} \Bigg|_{k=k_{\pm}(\phi)}$$

where  $W^{-1} \equiv \int dk/(2\pi)$  is the sample width and  $k_{\pm}(\phi)$  is obtained from  $\varepsilon(k_{\pm}(\phi)) = \mu$ .

## 4. Analytical Results for the Isotropic Rashba Model

### 4.1 Magnetization in the High-Density Regime (HDR)

For an electric field  $\mathbf{E} = E_x \hat{x}$ , the spin density along  $\hat{y}$  in the HDR (both bands occupied) is [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$M_y = \frac{\mu_B |e| \tau}{2\pi} m \alpha [\hat{z} \times \mathbf{E}]_y$$

**Key result:** In the HDR, the magnetization is **constant, independent of the Fermi energy**  $E_F$ , and proportional to the Rashba SOC parameter  $\alpha$  and the effective mass  $m$ .

### 4.2 Magnetization in the Low-Density Regime (LDR)

In the LDR (only the lower band occupied) [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$M_y = \frac{\mu_B |e| \tau}{2\pi} \sqrt{m^2 \alpha^2 + 2m E_F} [\hat{z} \times \mathbf{E}]_y$$

For small  $E_F$  near the band crossing, this can be expanded as:

$$M_y \approx \frac{\mu_B |e| \tau}{2\pi} \left( \alpha m + \frac{1}{2} \frac{E_F}{\alpha} \right) [\hat{z} \times \mathbf{E}]_y$$

The spin density **increases linearly with the Fermi energy**. In the limit  $\alpha \rightarrow 0$ , the second term vanishes because  $E_F$  has a quadratic dependence on  $\alpha$ .

### 4.3 Vector Form of the Edelstein Magnetization

The general vector form of the Edelstein magnetization for the isotropic Rashba model is:

$$\mathbf{M} = \chi_{EE} [\hat{\mathbf{z}} \times \mathbf{E}]$$

This means the magnetization is **in-plane and perpendicular** to the applied electric field. For example, if  $\mathbf{E} = E_x \hat{x}$ , then  $\mathbf{M} = M_y \hat{y}$ .

## 5. Anisotropic Rashba Model ( $C_{2v}$ Symmetry)

### 5.1 Anisotropic Hamiltonian

For systems with  $C_{2v}$  symmetry, the Rashba Hamiltonian generalizes to [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\hat{H} = \frac{\hbar^2 k_x^2}{2m_x} + \frac{\hbar^2 k_y^2}{2m_y} + \alpha_y k_y \hat{\sigma}_x - \alpha_x k_x \hat{\sigma}_y$$

with anisotropic effective masses  $m_x \neq m_y$  and anisotropic SOC parameters  $\alpha_x \neq \alpha_y$ .

### 5.2 Edelstein Susceptibility with Mass Anisotropy

In the HDR, the Edelstein susceptibility as a function of the mass ratio  $r_m = m_y/m_x$  is [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\frac{\chi_{xy}}{\chi_0}(r_m) = 4\pi m_x \alpha \frac{r_m}{1 + \sqrt{r_m}}$$

### 5.3 Edelstein Susceptibility with SOC Anisotropy

As a function of the SOC ratio  $r_\alpha = \alpha_y/\alpha_x$  [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\frac{\chi_{xy}}{\chi_0}(r_\alpha) = 4\pi m \alpha_x \frac{r_\alpha}{1 + r_\alpha}$$

#### Key results:

- When  $r_m < 1$  or  $r_\alpha < 1$ , the susceptibility is **lower** than in the isotropic case.
- When  $r_m > 1$  or  $r_\alpha > 1$ , the susceptibility **increases**.
- The susceptibility **saturates** for  $r_\alpha \gg 1$ .
- It is possible to **boost the Edelstein response** by engineering the anisotropy.

## 6. Dependence on Model Parameters

### 6.1 Dependence on Rashba SOC Strength $\alpha$

The Edelstein susceptibility **increases linearly with**  $\alpha$  in the HDR [Gaiardoni et al., arXiv:2503.20712 (2025)]:

$$\chi_{xy} \propto \alpha \quad (\text{HDR})$$

This is confirmed numerically and analytically. As  $\alpha$  increases, the value of  $\chi_{xy}/\chi_0$  at which it reaches a plateau also increases [Gaiardoni et al., arXiv:2503.20712 (2025), Fig. 3].

### 6.2 Dependence on Fermi Energy / Chemical Potential

- In the **HDR**:  $\chi_{xy}$  is **constant** as a function of  $E_F$  [Gaiardoni et al., arXiv:2503.20712 (2025)].
- In the **LDR**:  $\chi_{xy}$  **increases linearly** with  $E_F$  [Gaiardoni et al., arXiv:2503.20712 (2025), Eq. (10)].

### 6.3 Dependence on Chirality and Sign of $\alpha$

The sign of the Rashba parameter  $\alpha$  determines the **chirality** of the spin texture. For  $\alpha > 0$  and  $\alpha < 0$ , the spin orientations on the Fermi surfaces reverse [Auvray et al., 2018; Funato & Matsuo, arXiv:2107.03115 (2021)]. This flips the sign of the Edelstein magnetization.

The **sign of the effective mass**  $m^*$  also matters: it determines the splitting order of inner and outer energy branches ( $E_+$ ,  $E_-$ ) [Auvray et al., 2018].

### 6.4 Dependence on Fermi Velocity

Through the density of states and transport time, the Fermi velocity  $v_F$  enters the Edelstein susceptibility. The spin density can be expressed through the dimensionless Rashba parameter [Funato & Matsuo, arXiv:2107.03115 (2021)]:

$$\tilde{\alpha}_R = \frac{m\alpha}{k_F} = \frac{\alpha}{v_F}$$

Thus, for fixed  $\alpha$ , increasing the Fermi velocity (decreasing  $\tilde{\alpha}_R$ ) reduces the dimensionless Edelstein response.

## 7. Out-of-Plane Edelstein Effect in p-Wave Magnets

## 7.1 Two-Band Model with p-Wave and Rashba Terms

Ezawa [arXiv:2501.01888 (2025)] proposed a model combining a p-wave magnet with Rashba SOC:

$$H(\mathbf{k}) = \frac{\hbar^2(k_x^2 + k_y^2)}{2m}\sigma_0 + \lambda(k_x\sigma_y - k_y\sigma_x) + Jk_x \mathbf{n} \cdot \boldsymbol{\sigma}$$

where  $\lambda$  is the Rashba coupling,  $J$  is the p-wave Néel coupling, and  $\mathbf{n}$  is the p-wave Néel vector.

## 7.2 Out-of-Plane Néel Vector ( $\mathbf{n} = \hat{z}$ )

The energy spectrum is:

$$\varepsilon_{\pm} = \frac{\hbar^2 k^2}{2m} \pm k\sqrt{F(\phi)}, \quad F(\phi) \equiv \lambda^2 + J^2 \cos^2 \phi$$

The spin expectation values are:

$$S_x^{\pm} = \mp \frac{\lambda \sin \phi}{\sqrt{F(\phi)}}, \quad S_y^{\pm} = \pm \frac{\lambda \cos \phi}{\sqrt{F(\phi)}}, \quad S_z^{\pm} = \pm \frac{J \cos \phi}{\sqrt{F(\phi)}}$$

The Edelstein susceptibility components (up to first order in  $J$  and  $\lambda$  for  $\mu > 0$ ) are [Ezawa, arXiv:2501.01888 (2025)]:

$$\begin{aligned} \alpha_{xx}^{ME} &= 0, \\ \alpha_{yx}^{ME} &= -\frac{g\mu_B m}{2\pi\hbar^3 W} \lambda, \\ \alpha_{zx}^{ME} &= -\frac{g\mu_B m}{2\pi\hbar^3 W} J \end{aligned}$$

**Key result:** For an out-of-plane p-wave Néel vector, an in-plane electric field  $\mathbf{E} = E_x \hat{x}$  induces an **out-of-plane magnetization**  $M_z = e\tau\alpha_{zx}^{ME} E_x$  proportional to  $J$ , in addition to the in-plane component  $M_y$  proportional to  $\lambda$ .

- $\alpha_{zx}^{ME} = 0$  when  $J = 0$  or  $\lambda = 0$  (both are required for the out-of-plane effect).
- The sign of  $\alpha_{zx}^{ME}$  depends on the sign of  $J$  but not on  $\lambda$ .

## 7.3 In-Plane Néel Vector ( $\mathbf{n} = \hat{y}$ )

The Edelstein susceptibility is [Ezawa, arXiv:2501.01888 (2025)]:

$$\begin{aligned}\alpha_{yx}^{ME} &= -\frac{mg\mu_B}{4\pi\hbar^3W} (J + 2\lambda), \\ \alpha_{xx}^{ME} &= \alpha_{zx}^{ME} = 0\end{aligned}$$

## 8. Acoustic (Mechanical) Edelstein Effect

Funato & Matsuo [arXiv:2107.03115 (2021)] studied the *acoustic Edelstein effect* (AEE)—spin density induced by lattice distortion dynamics. The spin density from the Fermi surface term is:

$$\langle \hat{\sigma}_\alpha \rangle_{\text{surf}}^k = i\omega \alpha_R m \nu_0 \tilde{a}_2 \tau [\hat{z} \times \mathbf{u}_{q,\omega}]_\alpha$$

where  $\mathbf{u}_{q,\omega}$  is the lattice velocity field. The **current–spin conversion efficiency** for the AEE is:

$$\lambda_A = -\frac{\tilde{\alpha}_R}{k_F} \frac{e}{\mu} \left[ \frac{a_3}{\tilde{a}_2} - 2\tilde{\alpha}_R^2 \left( 1 - \frac{\tilde{a}_2}{2a_1} \right) \right]$$

while for the conventional electric EE:

$$\lambda_e = -\frac{\tilde{\alpha}_R}{k_F} \frac{e}{\mu} \left( 1 - \frac{a_1}{\tilde{a}_2} \right) \left[ \frac{a_3}{\tilde{a}_2} - 2\tilde{\alpha}_R \left( 2 - \frac{\tilde{a}_2}{a_1} - \frac{a_1}{\tilde{a}_2} \right) \right]$$

The AEE and conventional EE show **different  $\alpha_R$ -dependencies**, indicating they are distinct mechanisms [Funato & Matsuo, arXiv:2107.03115 (2021)].

## 9. Effect of Magnetic Field: Out-of-Plane Polarization

Engel, Rashba, and Halperin [arXiv:cond-mat/0609078 (2006)] showed that with parallel in-plane electric and magnetic fields, **out-of-plane spin polarization** can be generated for anisotropic impurity scattering or non-parabolic bands.

For the Rashba model with  $\mathbf{B} = B\hat{x}$  and  $\mathbf{E} = E\hat{x}$ , the spin-orbit field is:

$$\mathbf{b}(\mathbf{k}) = 2\alpha \hat{z} \times \mathbf{k} + \Delta_x \hat{x}, \quad \Delta_x = g^* \mu_B B$$

The **Bloch equations** for the total spin polarization are [Engel et al., arXiv:cond-mat/0609078 (2006)]:

$$\dot{\mathbf{s}} = \langle \mathbf{b} \rangle \times \mathbf{s} - \overleftrightarrow{\tau}_s^{-1} \mathbf{s} + \mathbf{\Gamma}$$

with generation rate:



$$\Gamma = \left( \frac{1}{2}\nu\Delta_x\tau_{xy}^{-1}, \frac{1}{2}\nu\langle b_y\rangle\tau_{xy}^{-1}, \frac{1}{4}\nu\langle b_y\rangle\Delta_x\tilde{\gamma}_0 \right)$$

The **static (dc) out-of-plane polarization** is:

$$s_z = \frac{1}{2}\nu\alpha eE_x\tau \frac{\Delta_x\tau_z}{1 + \Delta_x^2\tau_{xy}\tau_z} \tilde{\gamma}_0$$

where  $\tilde{\gamma}_0$  characterizes the anisotropy of impurity scattering and/or band non-parabolicity ( $\tilde{\gamma}_0 = \zeta + 3$  for small-angle scattering).

**Key result:** The out-of-plane polarization  $s_z$  is **proportional to  $\Delta_x \propto B$** , so it reverses sign with the magnetic field. It vanishes for isotropic scattering and parabolic bands ( $\tilde{\gamma}_0 = 0$ ).

## 10. Numerical Implementation Guidelines

### 10.1 Algorithm for Computing the Edelstein Magnetization

1. **Define parameters:**  $m, \alpha, E_F$  (or  $\mu$ ),  $\tau, T$ , and grid in  $\mathbf{k}$ -space.
2. **Compute band energies:**  $\varepsilon_{\pm}(\mathbf{k}) = \frac{\hbar^2 k^2}{2m} \pm \alpha \hbar k$ .
3. **Compute spin expectation values** on each band:  $\langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^{\pm} = (\pm \sin \theta, \mp \cos \theta, 0)$ .
4. **Compute group velocities:**  $\mathbf{v}^{\pm}(\mathbf{k}) = \nabla_{\mathbf{k}} \varepsilon^{\pm}(\mathbf{k})$ .
5. **Apply the linear-response formula:**

$$\mathbf{M} = -\mu_B \sum_{\mathbf{k}, \nu} |e| \bar{\tau}_{\mathbf{k}}^{\nu} (\mathbf{v}^{\nu} \cdot \mathbf{E}) \delta(\varepsilon_{\mathbf{k}}^{\nu} - E_F) \langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^{\nu}$$

or the equivalent zero-temperature form:

$$\mathbf{M} = e\tau g\mu_B \int \frac{d^2k}{(2\pi)^2} \mathbf{S}(\mathbf{k}) (\mathbf{E} \cdot \mathbf{v}) \delta(\varepsilon_{\mathbf{k}} - \mu)$$

6. **Extract susceptibility tensor:**  $\chi_{ij} = \partial M_i / \partial E_j$ .

### 10.2 Python Code Skeleton

```

import numpy as np

# Parameters
m = 0.152          # effective mass in eV-1 Å-2 (oxide interfaces)
alpha = 52.0       # Rashba SOC in meV·Å
EF = 60.0          # Fermi energy in meV
tau = 1e-12        # transport time in seconds
mu_B = 5.788e-5    # Bohr magneton in eV/T
e = 1.602e-19      # electron charge (C)
hbar = 6.582e-16   # hbar in eV·s

def dispersion(k, m, alpha):
    """Rashba energy dispersion."""
    return hbar**2 * k**2 / (2*m) + np.array([-1, 1]) * alpha * k

def spin_expectation(theta):
    """Spin expectation values for the two chiral bands."""
    return np.array([np.sin(theta), -np.cos(theta), 0.0]), \
           np.array([-np.sin(theta), np.cos(theta), 0.0])

def group_velocity(k, theta, m, alpha):
    """Group velocities for the two bands."""
    vx = hbar*k*np.cos(theta)/m + np.array([-alpha, alpha])*np.cos(theta)
    vy = hbar*k*np.sin(theta)/m + np.array([-alpha, alpha])*np.sin(theta)
    return vx, vy

# Build k-grid in polar coordinates
Nk, Ntheta = 500, 500
kmax = 0.5 # in units of inverse Angstrom
k = np.linspace(1e-6, kmax, Nk)
theta = np.linspace(0, 2*np.pi, Ntheta)

# Compute magnetization for E along x
Mx, My = 0.0, 0.0
for ki in k:
    eps_plus, eps_minus = dispersion(ki, m, alpha)
    for tj in theta:
        # Band + (helicity +1)
        vx_plus, vy_plus = group_velocity(ki, tj, m, alpha)
        s_plus = spin_expectation(tj)[0]
        delta = abs(eps_plus - EF) < 1e-3 # delta-function approximation
        My += -mu_B*e*tau*(vx_plus*Ex)*delta*s_plus[1]*ki

# Normalize by area factors...

```

## 10.3 Expected Graphics

The following explicit plots should be produced:

1. **Spin density  $M_y$  vs. electric field magnitude  $|E|$**  — linear relationship (since we are in the linear-response regime).
2. **Spin density components vs. electric field direction** —  $M_x = \chi_{xx}E_x + \chi_{xy}E_y$ ; for the isotropic Rashba model,  $\chi_{xx} = 0$ ,  $\chi_{xy} = -\chi_{yx}$ , so  $\mathbf{M}$  is always perpendicular to  $\mathbf{E}$ .
3.  **$M_y$  vs.  $\alpha$**  — linear increase in HDR [Gaiardoni et al., Fig. 3, right panel].
4.  **$M_y$  vs.  $E_F$  (or  $\mu$ )** — plateau in HDR, linear increase in LDR [Gaiardoni et al., Fig. 2, left panel; Fig. 3, left panel].
5.  **$M_y$  vs. anisotropy ratios  $r_m, r_\alpha$**  — increasing function saturating for large  $r_\alpha$  [Gaiardoni et al., Fig. 6].
6. **Out-of-plane component  $M_z$  vs.  $J$  (p-wave parameter)** — linear in  $J$ , zero for  $J = 0$  [Ezawa, Fig. 2(a1)].
7.  **$M_z$  vs.  $\lambda$  (Rashba)** — for the p-wave model with out-of-plane Néel vector,  $M_z \propto J$  (independent of  $\lambda$  for small  $\lambda$ ) [Ezawa, Fig. 2(b1)].
8.  **$M_z$  vs. magnetic field  $B$**  (for anisotropic scattering) — linear in  $B$  near  $B = 0$ , with Hanle-type saturation [Engel et al., Eq. (16)].

## 11. Summary of Key Formulas for the Model

| Quantity           | Formula                                                                                          | Source                            |
|--------------------|--------------------------------------------------------------------------------------------------|-----------------------------------|
| Rashba Hamiltonian | $\hat{H} = \frac{p^2}{2m} + \alpha \hat{z} \cdot (\mathbf{p} \times \boldsymbol{\sigma})$        | Bychkov & Rashba (1984)           |
| Band energies      | $\varepsilon_{\pm} = \frac{\hbar^2 k^2}{2m} \pm \alpha \hbar k$                                  | Gaiardoni et al. (2025)           |
| Spin texture       | $\langle \boldsymbol{\sigma} \rangle_{\mathbf{k}}^{\pm} = (\pm \sin \theta, \mp \cos \theta, 0)$ | Gaiardoni et al. (2025)           |
| HDR magnetization  | $M_y = \frac{\mu_B  e  \tau}{2\pi} m \alpha [\hat{z} \times \mathbf{E}]_y$                       | Gaiardoni et al. (2025), Eq. (8)  |
| LDR magnetization  | $M_y = \frac{\mu_B  e  \tau}{2\pi} \sqrt{m^2 \alpha^2 + 2m E_F} [\hat{z} \times \mathbf{E}]_y$   | Gaiardoni et al. (2025), Eq. (9)  |
| Anisotropic (mass) | $\chi_{xy}/\chi_0 = 4\pi m_x \alpha r_m / (1 + \sqrt{r_m})$                                      | Gaiardoni et al. (2025), Eq. (12) |
| Anisotropic (SOC)  | $\chi_{xy}/\chi_0 = 4\pi m \alpha_x r_\alpha / (1 + r_\alpha)$                                   | Gaiardoni et al. (2025),          |

| Quantity                  | Formula                                                                                                          | Source                        |
|---------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------|
|                           |                                                                                                                  | Eq. (12)                      |
| Out-of-plane<br>(p-wave)  | $\alpha_{zx}^{ME} = -\frac{g\mu_B m}{2\pi\hbar^3 W} J$                                                           | Ezawa (2025), Eq. (13)        |
| Out-of-plane<br>(B-field) | $s_z = \frac{1}{2}\nu\alpha e E_x \tau \frac{\Delta_x \tau_z}{1 + \Delta_x^2 \tau_{xy} \tau_z} \tilde{\gamma}_0$ | Engel et al. (2006), Eq. (16) |

## 12. References

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